BOUNDED CONVERGENCE OF THE HYPERCOHOMOLOGY SPECTRAL SEQUENCE ON QUASI-COMPACT QUASI-SEPARATED SCHEMES

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Allowed standard prerequisites (editorial).

- The hypercohomology spectral sequence arises from the filtered-object construction applied to the canonical truncation filtration of K. invoked at proof lines 782-784. Reduces convergence to vanishing and agreement statements for the truncations.
- On a quasi-compact quasi-separated scheme the cohomology of a suitable truncation vanishes in large degree of the filtration index, by the nice-K-injective application with the global sections functor and the quasi-coherence bound on higher direct images. invoked at proof lines 793-796. Gives the first requirement H^n(X, tau_leq_-p K) = 0 for p » 0.
- For a general ringed space, the canonical truncation does not change H^n once -p <= n. used at proof lines 797-799. Gives the second requirement for p « 0.

Source reference. Original remark 0BKM: sources/stacks-chow/tags.txt, source lines 14565–14565. Cited at statement lines 777-778 as the spectral sequence whose bounded convergence is the claim; the proof of convergence is inside the counted span.

Source reference. Original lemma 0BKK: sources/stacks-chow/tags.txt, source lines 14563–14563. Cited at proof lines 782-784 for the reduction to truncation statements; the vanishing and agreement arguments are inside the counted span.

Source reference. Original lemma 08D4: sources/stacks-perfect/pinned-original-source.tex, source lines 375–375. Cited at proof lines 793-794 for the vanishing requirement; the application with Gamma(X, -) and the bound is inside the counted span.

Source reference. Original lemma 01XJ: sources/stacks-chow/tags.txt, source lines 2520–2520. Cited at proof lines 795-796 for the bound that makes the vanishing effective; the conclusion is inside the counted span.

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Lemma 0.2. *Let X be a quasi-separated and quasi-compact scheme. For any object K of $D_{QCoh}(\mathcal{O}_X)$ the spectral sequence*

$$E_2^{i,j} = H^i(X, H^j(K)) \Rightarrow H^{i+j}(X, K)$$

of Cohomology, Example 0BKM is bounded and converges.

Proof. By the construction of the spectral sequence via Cohomology, Lemma 0BKK using the filtration given by $\tau_{\leq -p}K$, we see that suffices to show that given $n \in \mathbf{Z}$ we have

$$H^n(X, \tau_{\leq -p}K) = 0 \text{ for } p \gg 0$$

and

$$H^n(X, K) = H^n(X, \tau_{\leq -p}K) \text{ for } p \ll 0$$

The first follows from Lemma 08D4 applied with $F = \Gamma(X, -)$ and the bound in Cohomology of Schemes, Lemma 01XJ. The second holds whenever $-p \leq n$ for any ringed space $(X, \mathcal{O}_X)$ and any $K \in D(\mathcal{O}_X)$. □